

Experimental Twin-Field Quantum Key Distribution through Sending or Not Sending

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Channel loss seems to be the most severe limitation on the practical application of long distance quantum key distribution. The idea of twin-field quantum key distribution can improve the key rate from the linear scale of channel loss in the traditional decoy-state method to the square root scale of the channel transmittance. However, the technical demands are rather tough because they require single photon level interference of two remote independent lasers. Here, we adopt the technology developed in the frequency and time transfer to lock two independent laser wavelengths and utilize additional phase reference light to estimate and compensate the fiber fluctuation. Further, with a single photon detector with a high detection rate, we demonstrate twin field quantum key distribution through the sending-or-not-sending protocol with a realistic phase drift over 300 km optical fiber spools. We calculate the secure key rates with the finite size effect. The secure key rate at 300 km (1.96×10^{-6}) is higher than that of the repeaterless secret key capacity (8.64×10^{-7}).

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Introduction.—Although quantum key distribution (QKD) can, in principle, offer secure private communication [1–7], there are still some technical limitations on practical long distance quantum communication. Perhaps the most severe of these is channel loss, given that quantum signals cannot be amplified. Much effort has been made towards QKD over a long distance. Theoretically, the decoy-state method [8–10] can improve the key rate of coherent-state based QKD from scaling quadratically to linearly with the channel transmittance, as what behaves of a perfect single-photon source. This method can defeat the photon-number-splitting attack to the imperfect source, and the coherent state is used as if only the single-photon pulses were used for key distillation, and hence, it can reach a key rate in the linear scale of channel loss, as the perfect single-photon source does.

Remarkable theoretical progress was made toward achieving practical, secure QKD over longer distance with the proposal of twin-field QKD (TFQKD) [11], which improves the key rate scaling to follow the square root of the channel transmittance. It shows that the coherent-state source can actually be an advantage over the single-photon source because the postselection of the phase coherence of the twin fields from Alice and Bob can potentially lead to secure QKD

with the encoding state of a single photon and vacuum, and their linear superpositions. This method has the potential to achieve a key rate that scales with the square root of channel transmittance and can, by far, break the known distance limit of existing protocols in practical QKD [12–20]. The theoretical secure key rate can be even higher than the repeaterless secret key capacities, known as the Takeoka-Guha-Wilde bound [19] and the Pirandola-Laurenza-Ottaviani-Bianchi (PLOB) bound [20]. However, considerable work still remains to make this a reality.

First, there is the theoretical challenge of combining the phase information postselection with the traditional decoy-state method. Second, it is technically demanding of precise single photon interference over long distance. Towards this goal, the sending-or-not-sending (SNS) protocol [21] was proposed. This involves small sending probabilities for both Alice and Bob and then uses sending and not-sending decisions for the bit value encoding in the Z basis with the effective heralding events announced by Charlie. In this way, as was shown in [21], one can continue to use the tagged model and the conventional decoy-state method in the protocol. In addition, since the protocol encodes the bit values using the almost error-free Z basis, it can tolerate a high error rate in the X basis.

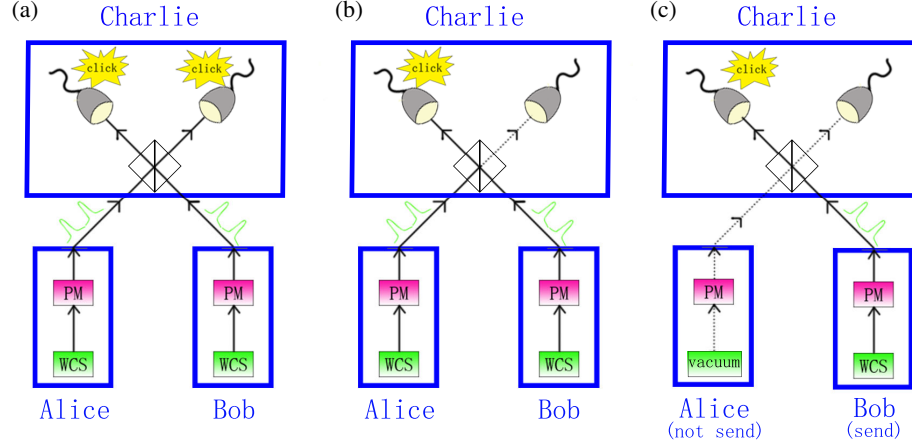


FIG. 1. Schematics for three different protocols. (a) Decoy-state MDIQKD, where coherent-state pulse pairs in BB84 encoding are sent out and effective events are heralded by twofold clicking. The key rate scales linearly with the channel transmittance. (b) The original decoy state TFQKD [11], where twin fields of coherent states with random phase shifts are sent out in the X and Y bases, and the effective events are heralded by single clicking. The key rate scales with the square root of the channel transmittance. Single-photon interference from remote independent sources are needed for both bases. There can be misalignment errors in both bases and the post announced phase-shift information renders the decoy-state method invalid. (c) Decoy-state SNSTFQKD [21]. In the Z basis, each side independently decides sending with a small probability. The events that one side decides sending, the other side decides not sending, and one and only one detector clicks (as shown in the figure) are the target events to generate secure keys. It is fault tolerant to a large misalignment error in the X basis because there is no misalignment error in the Z basis. The traditional decoy-state method holds because the phase-shift information in the Z basis is never announced. The heralding of a single clicking makes the effective events in the Z basis, and the key rate is in the scale of square root of the channel transmittance. WCS: weak coherent source.

Here, we report an experimental demonstration of twin field QKD through SNS protocol (SNSTFQKD) over optical fiber spools.

Protocol.—Consider the schematic of the SNSTFQKD protocol [21] shown in Fig. 1. Here, we implement the protocol using a practical four-intensity method [22], where each party exploits four different intensities, namely, 0, μ_1 , μ_2 , and μ_z .

Alice and Bob randomly choose the X or Z basis with probabilities p_X and $1 - p_X$, respectively. In the X basis, both Alice and Bob prepare and send decoy pulses. Phase shifts of θ_A and θ_B are privately applying to their pulses. A Z -basis event is regarded as being effective if Charlie announces that only one detector clicked. For an X -basis event to be effective, we need an extra phase-slice condition to reduce the observed error rate in the basis. Without a reasonable phase-slice condition, the observed X -basis error rate can be too much higher than the actual phase-flip error rate in the Z basis. Note that Charlie does not have to be honest, and whatever he announces does not undermine the security. But if Charlie wants to make a good key rate, he will have to try to make a faithful announcement on everything.

An error in the X basis is defined as Charlie announcing a click of the right (left) detector associated with an effective event in the X basis where the phase difference between the pulse pair from Alice and Bob would probably cause a left (right) clicking at Charlie's measurement setup. A Z -basis effective event that Alice (Bob) has decided to

send and Bob (Alice) has decided not to send corresponds to a bit value 1 (0). The values of e_1^{ph} and s_1 , the yield of single-photon effective events in the Z basis, can be calculated by the conventional decoy-state method [21,22]. See Supplemental Material [23] for detailed calculations.

Experiment.—The experimental layout is shown in Fig. 2(a). In Alice's and Bob's labs, independent continuous wave (cw) lasers are used as light sources. The light is modulated to 16 different phases with a phase modulator (PM) and encoded with three amplitude modulators (AMs). In the experiment, we set the basic period to 5 μs , during which 100 signal pulses are sent in the first 3 μs with a 2 ns pulse width and a 30 ns interval, followed by 4 phase reference pulses in the next 1.2 μs , to estimate the relative phase between Alice's and Bob's channels, and with a final 0.8 μs vacuum state as the superconducting nanowire single-photon detectors (SNSPDs) recovery time. The signal intensities are set to the optimized decoy states μ_z , ν_1 , ν_2 , or 0 (see Supplemental Material [23] for details about the encoding).

Then, the signals are transmitted from Alice and Bob to Charlie, where they interfere. Since interference requires identical input signals, polarization controllers (PCs) and polarization beam splitters (PBSs) before the polarization maintaining beam splitters (BSs) are necessary to compensate for the polarization drift of the channel. The interference results are then detected with SNSPDs and recorded with a high speed time tagger.

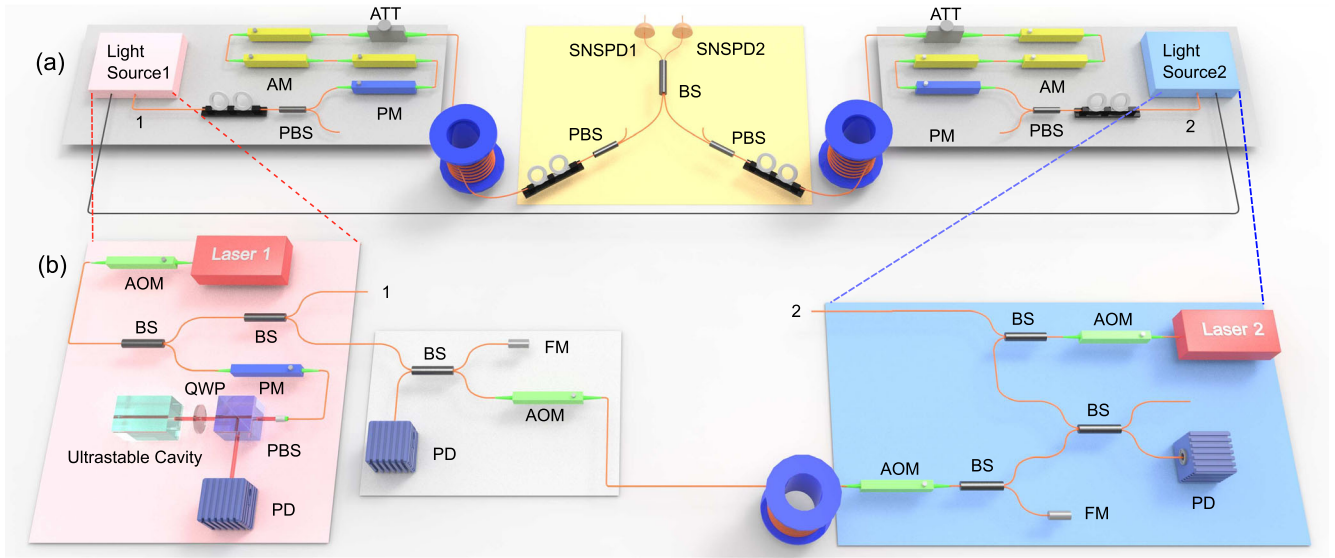


FIG. 2. (a) Schematic of our experimental setup. Alice and Bob use frequency-locked continuous-wave (cw) lasers as sources. These lasers are then modulated by a phase modulator (PM) and three amplitude modulators (AMs) for phase randomization, encoding, and decoy intensity modulation. The pulses are then attenuated by an attenuator (ATT) and sent out via fiber spools to Charlie. At Charlie's measurement station, the pulses from Alice and Bob pass through polarization controllers (PCs) and polarization beam splitters (PBSs), then interfere at a beam splitter (BS). Finally, the light is measured by superconducting nanowire single-photon detectors (SNSPDs). (b) Frequency-locking system for Alice's and Bob's lasers. The fiber length between Alice and Bob is set to be the same as the total length of the signal fiber. AOM: acousto-optic modulator, FM: Faraday mirror, PD: photodiode. QWP: Quarter-Wave Plate.

The main technical challenge in realizing SNSTFQKD is to control the twin fields' phase evolution. As was pointed out in [11], the differential phase fluctuation between the two users can be written as

$$\delta_{ba} = \frac{2\pi}{s} (\Delta\nu L + \nu\Delta L), \quad (1)$$

where ν is the optics frequency of the light, L is the length of the fiber, s is the speed of light in the fiber. Thus, two sources contributing to the phase difference should be compensated: the first term in the equation indicates the frequency difference between Alice and Bob; the second term stands for the phase drift in the fiber. As an example, the measured phase drift rate follows a Gaussian distribution with a standard deviation of 7.4 rad ms^{-1} for 150 km total fiber distance. The drift is a bit larger than reported in [11] mainly due to the relatively noisy environment in our lab (See Supplemental Material [23] for details).

To deal with the phase difference caused by the wavelength difference, we adopt a frequency-locking method, as shown in Fig. 2(b). In Alice's lab, a continuous wave laser with a center wavelength of 1550.12 nm and a line width of several kilohertz is used as the seed laser. The seed laser is locked to a 10-cm-long ultrastable cavity with a finesse around 250 000, using the Pound-Drever-Hall technique [24,25], to suppress its linewidth from a few kilohertz to approximately ten hertz. The light is then split into two parts, one is used as Alice's source and another to lock Bob's optical frequency. This locking beam is further split

into two parts, one is reflected by a Faraday mirror (FM) as the local reference, while the other is frequency-modulated by an acousto-optic modulator (AOM) and sent to Bob. Here, the fiber length is set to be the same as the signal transmission distance to demonstrate the system's practicability.

To ensure Bob receives light with the same frequency as Alice, we need to compensate for any phase noise in the transmission fiber. On Bob's side, a BS and a FM work as a partial reflector for the locking light. A second AOM with a fixed frequency shift is inserted before the BS, to distinguish the reflected laser light from backscattering and facet reflection noise. The whole setup, including Alice's and Bob's reflectors, is actually a large Michelson interferometer. Thus, the phase error signal can be generated by mixing the beat pattern at Alice's photodiode (PD) with a fixed-frequency reference. A PI servo then generates the feedback signal for Alice's AOM, compensating for the phase drift on Bob's side. Bob then locks his own laser to the transmitted one with fast AOM feedback and slow piezoelectric feedback with a large adjustment range. The whole locking system can work continuously for weeks.

To handle the phase difference caused by path fiber noise, we use phase reference pulses to estimate it. During the phase reference time, Alice adjusts the phases to $\theta_A = \{0, \pi/2, \pi \text{ and } 3\pi/2\}$ for the four types of pulses, while Bob maintains his phase at 0. So the total phase difference before the interference is $\phi = \theta_A - \theta_B + \Delta\varphi_T$, where $\Delta\varphi_T$ is the relative phase difference of the fiber channel. Thus, Charlie records the interference detections for four different

phases ϕ , and then uses these results to search for the most probable phase difference $\Delta\varphi_T$ using a least squares method (see Supplemental Material [23] for further details of the phase estimation process).

To accumulate enough detections to estimate the relative phase, we set the intensity of the phase reference pulses such that a total of approximately 20 detection events occurred in the $1.2\ \mu\text{s}$ interval, which requires a peak counting rate of more than 40 MHz. This need for such a peak counting rate, together with the low dark count noise of less than 1000 Hz, imposes stringent requirements on single photon detectors. In particular, the low dark count rate must be achieved within a recovery time of only a few hundred of nanoseconds, making the problem even harder.

We developed SNSPDs with a two-parallel-nanowire serial-connected configuration [26] which cover an active area $16\ \mu\text{m}$ in diameter. The SNSPD's recovery time is mainly limited by its kinetic inductance [27]. This parallel configuration effectively reduces the kinetic inductance without scarifying detection efficiency. In addition, we inserted a 50 ohm shunt resistor between the dc arm of the bias tee and the ground at room temperature [28] to prevent the detector latching at high count rate.

Instead of actively stabilizing the relative phase between Alice and Bob, we compensate for the phase difference via postprocessing. Defining the estimated relative phase between Alice's and Bob's fibers as $\Delta\varphi_T$, we calculate the quantum bit error rate in the X basis for the detections that lie within the range

$$1 - |\cos(\theta_A - \theta_B + \Delta\varphi_T)| < \Lambda, \quad (2)$$

where θ_A (θ_B) is the random phase Alice (Bob) modulates on the signal and Λ is a preset range.

Then, we can calculate the secure key rate with a finite data size effect by the following formula [29] (See Supplemental Material [23] for further details of the protocol):

$$R = (1 - p_X)^2 \{2p_Z(1 - p_Z)a_1s_1[1 - H(e_1^{\text{ph}})] - fS_ZH(E_Z)\} - \frac{1}{N_{\text{total}}} \log_2 \frac{1}{\epsilon^5}, \quad (3)$$

where R is the final key rate, $a_1 = \mu_z e^{-\mu_z}$, s_1 is the yield of effective single-photon events in the Z basis, e_1^{ph} is the phase-flip error rate for the events in Z basis, S_Z and E_Z are the observed yield and bit-flip error rate for the Z basis, N_{total} is the total number of signal pulses sent, and $\epsilon = 10^{-10}$, corresponding to a total failure probability of 2×10^{-9} . The key rate would be even higher if we only considered the statistical fluctuation alone. Here, we assume an error correction efficiency of $f = 1.1$.

We tested SNSTFQKD with total fiber distances from 0 to 300 km between Alice and Bob. The detailed parameters are summarized in the Supplemental Material [23],

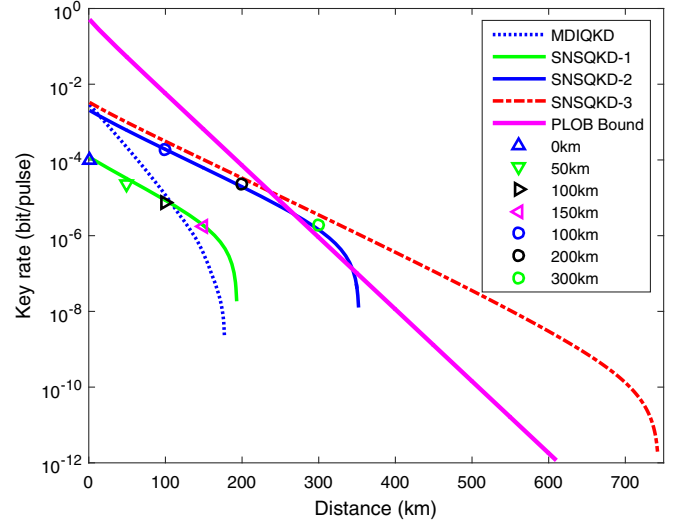


FIG. 3. Secure key rates and SNSTFQKD simulation results. The triangles show the experimental results for the first experimental test while the solid green curve represents the simulation results with a dark count probability of around 10^{-6} and X -basis baseline error of approximately 10%. For comparison, the dotted blue curve gives the result of simulating the four-intensity decoy-state MDIQKD protocol [16] with the same parameters but with 2% optical errors in the X basis. The circles show the experimental results for the second test while the solid blue curve represents the simulation with a dark count probability of around 10^{-7} and X -basis baseline error of approximately 2%. The total pulses sent by Alice and Bob for all the experimental tests are 7.2×10^{11} . The dotted-dashed red curve further assumes a total of 10^{14} pulses sent by Alice and Bob with 2% X -basis baseline error. Finally, the solid magenta line illustrates the PLOB bound [20].

including the optical efficiencies of the fibers and optical element, the intensities and proportions of the signal, and decoy states for each fiber length. In all the experiments with different fiber lengths, the total number of pulses sent by Alice and Bob is set to 7.2×10^{11} . The valid detections are 6.5×10^9 , 2.3×10^9 , 7.6×10^8 , and 2.5×10^8 for 0, 50, 100, and 150 km in the first experimental test, and 1.7×10^9 , 1.9×10^8 , and 2.4×10^7 for 100, 200, and 300 km in the second test.

The experimental result is summarized in Fig. 3. First, we experimentally tested SNSTFQKD at a dark count probability of 10^{-6} (equivalent to 1000 Hz), X -basis error rate of 10%. The secure key rate at 150 km is 1.72×10^{-6} per pulse, already higher than a simulated secure key rate of the measurement device independent quantum key distribution (MDIQKD) protocol [16] using the same parameter as in the experiment, but assuming lower (2%) optical errors in the X basis [17]. In fact, the simulation shows that the secure key rate already exceeds that of MDIQKD at 108 km.

Next, we reduced the dark count probability to approximately 10^{-7} (equivalent to 100 Hz) by upgrading the SNSPD to integrate an on-chip bandpass filter [30] inside,

and reduced the X -basis error rate to around 2% by using a linear amplifier to drive the modulators. The secure key rate at 300 km fiber distance is 1.96×10^{-6} , which is higher than the PLOB bound [20] of 8.64×10^{-7} per pulse. The simulation shows that SNSTFQKD breaks the bound at 267 km, and the transmission distance can be more than 350 km with the experimental parameters.

Finally, we simulate the secure key rate assuming a total of 10^{14} pulses are sent (with 2.6×10^5 valid detections accumulated at 720 km), and the single photon detector dark count probability reduced to 10^{-11} (equivalent to 0.1 Hz with 100 ps pulse width) [30]. All other parameters are set to that in the 300 km experiment. The simulation shows a maximum distribution distance of 742 km, and the SNSTFQKD protocol achieves a key rate above the PLOB bound [20] when the fiber distance is greater than 236 km.

In conclusion, we have developed phase locking and phase compensation technologies, tested the SNSTFQKD protocol experimentally, and demonstrated the generation of secure keys at fiber distances of up to 300 km, yielding a higher key rate than the repeaterless secret key capacity. The key rate calculation has fully considered finite size effects, thus, guaranteeing the security in a practical situation. We note that both the distance and key rate can be significantly improved by using two-way classical communication [31]. The experimental result also shows the SNSTFQKD protocol is robust against the phase mismatching, which is an important advantage in practice. The phase locking method used in the experiment has been found to be stable at a fiber distance of 1800 km [32], and the intensities of the phase reference pulses were within a few microwatts even at 1000 km. With currently available technology and the results of theoretical simulations with practical parameters, we expect that distribution distances of more than 500 km will be achieved in the near future.

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Note added.—Recently, we become aware of related works [33–35].

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